

ASSIGNMENT 11.1

Data Structures with AI: Implementing Fundamental Structures

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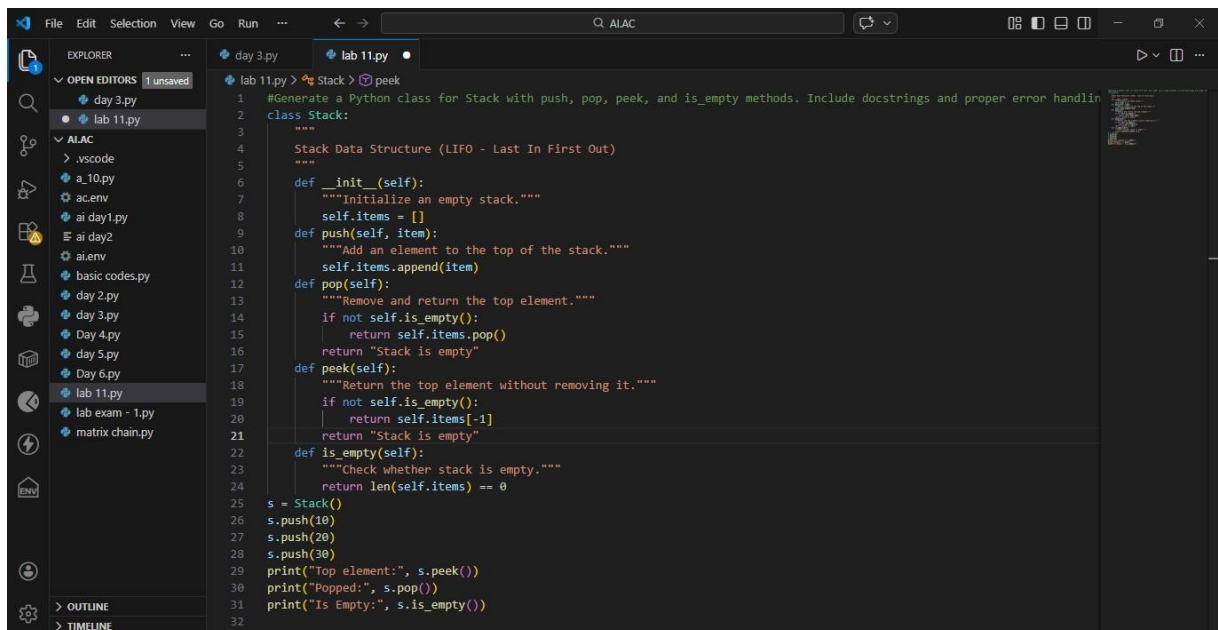
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Batch-51

Task 1: Stack Implementation

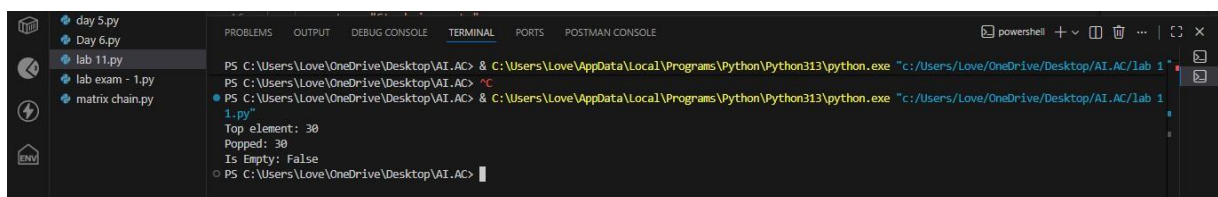
Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Prompt: Generate a Python class for Stack with push, pop, peek, and is_empty methods. Include docstrings and proper error handling.



```
1 #Generate a Python class for Stack with push, pop, peek, and is_empty methods. Include docstrings and proper error handling
2 class Stack:
3     """
4     Stack Data Structure (LIFO - Last In First Out)
5     """
6     def __init__(self):
7         """Initialize an empty stack."""
8         self.items = []
9     def push(self, item):
10        """Add an element to the top of the stack."""
11        self.items.append(item)
12    def pop(self):
13        """Remove and return the top element."""
14        if not self.is_empty():
15            return self.items.pop()
16        return "Stack is empty"
17    def peek(self):
18        """Return the top element without removing it."""
19        if not self.is_empty():
20            return self.items[-1]
21        return "Stack is empty"
22    def is_empty(self):
23        """Check whether stack is empty."""
24        return len(self.items) == 0
25
26 s = Stack()
27 s.push(10)
28 s.push(20)
29 s.push(30)
30 print("Top element:", s.peek())
31 print("Popped:", s.pop())
32 print("Is Empty:", s.is_empty())
```

OUTPUT:



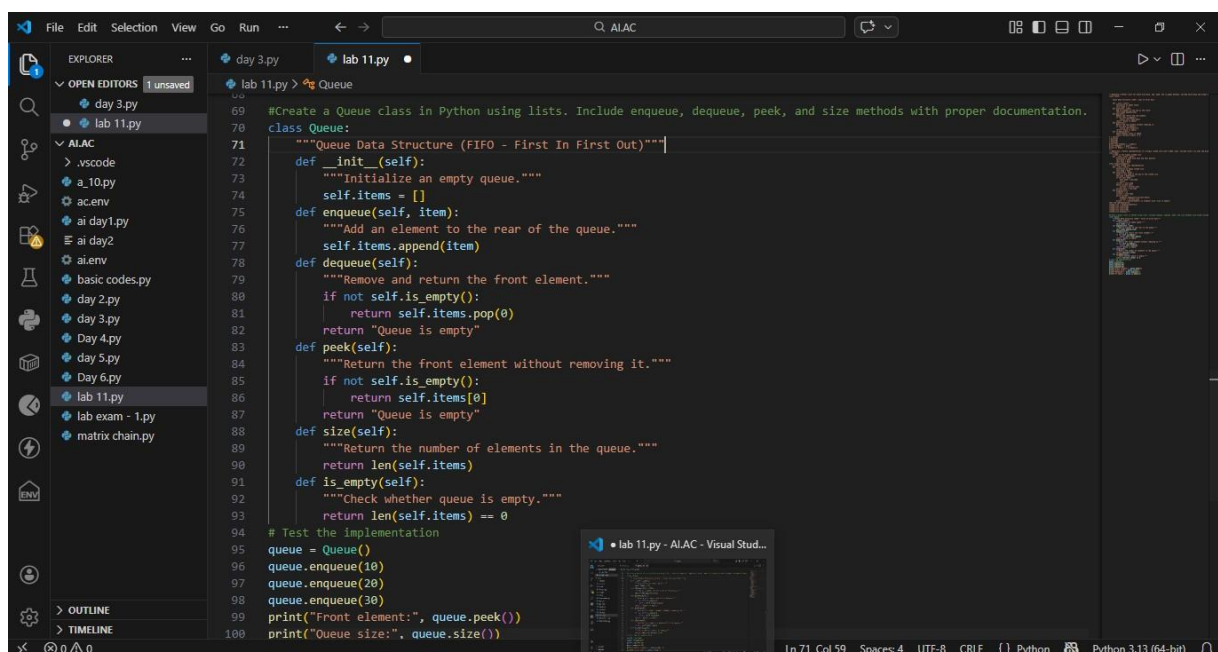
```
PS C:\Users\Love\OneDrive\Desktop\AI.AC> & C:\Users\Love\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Love/OneDrive/Desktop/AI.AC/lab 1"
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
PS C:\Users\Love\OneDrive\Desktop\AI.AC> & C:\Users\Love\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Love/OneDrive/Desktop/AI.AC/lab 1.py"
Top element: 30
Popped: 30
Is Empty: False
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Stack is a linear data structure that follows the LIFO (Last In First Out) principle, where the last element inserted is the first one removed. Operations such as push, pop, and peek are performed at one end called the top. It is commonly used in function calls, undo operations, and expression evaluation.

Task Description #2: Queue Implementation

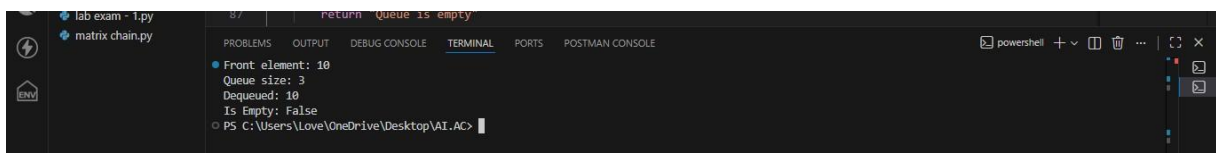
Task: Use AI to implement a Queue using Python lists.

Prompt: Create a Queue class in Python using lists. Include enqueue, dequeue, peek, and size methods with proper documentation.



```
69 #Create a Queue class in Python using lists. Include enqueue, dequeue, peek, and size methods with proper documentation.
70 class Queue:
71     """Queue Data Structure (FIFO - First In First Out)"""
72     def __init__(self):
73         """Initialize an empty queue."""
74         self.items = []
75     def enqueue(self, item):
76         """Add an element to the rear of the queue."""
77         self.items.append(item)
78     def dequeue(self):
79         """Remove and return the front element."""
80         if not self.is_empty():
81             return self.items.pop(0)
82         return "Queue is empty"
83     def peek(self):
84         """Return the front element without removing it."""
85         if not self.is_empty():
86             return self.items[0]
87         return "Queue is empty"
88     def size(self):
89         """Return the number of elements in the queue."""
90         return len(self.items)
91     def is_empty(self):
92         """Check whether queue is empty."""
93         return len(self.items) == 0
94
95 # Test the implementation
96 queue = Queue()
97 queue.enqueue(10)
98 queue.enqueue(20)
99 queue.enqueue(30)
100 print("Front element:", queue.peek())
101 print("Queue size:", queue.size())
```

OUTPUT:



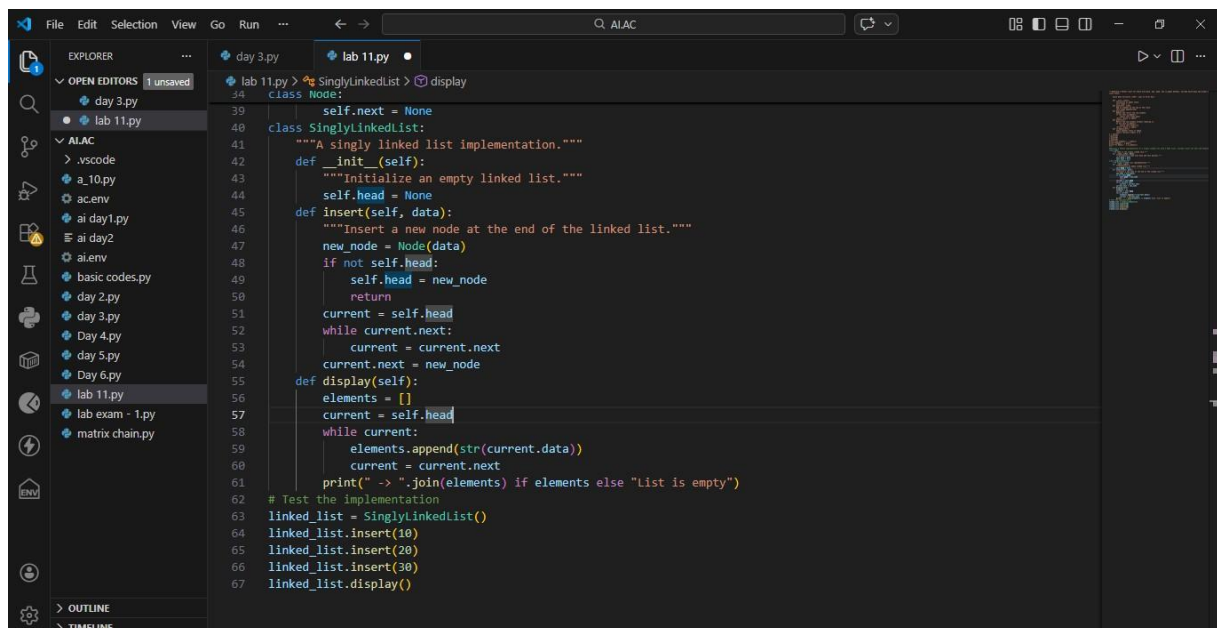
```
07 return "Queue is empty"
Front element: 10
Queue size: 3
Dequeued: 10
Is Empty: False
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Queue is a linear data structure that follows the FIFO (First In First Out) principle. This means the first element inserted is the first one removed.

Task Description #3: Linked List

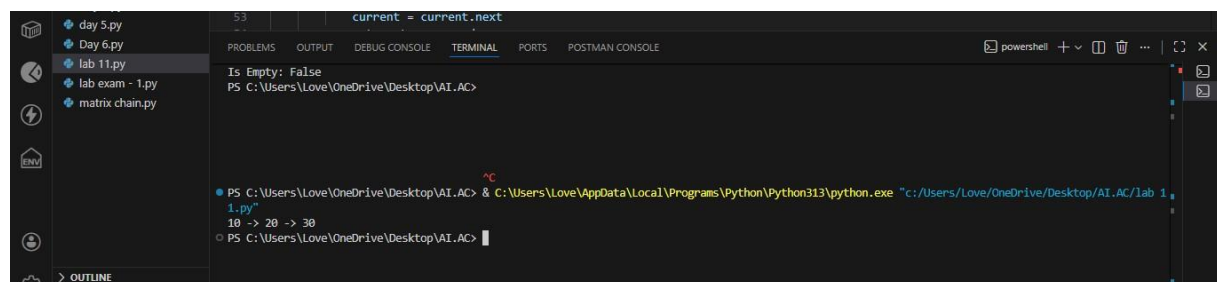
Task: Use AI to generate a Singly Linked List with insert and display methods.

Prompt : Generate a Python implementation of a Singly Linked List with a Node class. Include insert (at end) and display methods with docstrings.



```
34 class Node:
35     def __init__(self, data):
36         self.data = data
37         self.next = None
38
39 class SinglyLinkedList:
40     """A singly linked list implementation."""
41     def __init__(self):
42         """Initialize an empty linked list."""
43         self.head = None
44
45     def insert(self, data):
46         """Insert a new node at the end of the linked list."""
47         new_node = Node(data)
48         if not self.head:
49             self.head = new_node
50             return
51         current = self.head
52         while current.next:
53             current = current.next
54         current.next = new_node
55
56     def display(self):
57         elements = []
58         current = self.head
59         while current:
60             elements.append(str(current.data))
61             current = current.next
62         print(" -> ".join(elements) if elements else "List is empty")
63
64 # Test the implementation
65 linked_list = SinglyLinkedList()
66 linked_list.insert(10)
67 linked_list.insert(20)
68 linked_list.insert(30)
69 linked_list.display()
```

OUTPUT:



```
53 current = current.next
54
55 Is Empty: False
56 PS C:\Users\Love\OneDrive\Desktop\AI.AC>
57
58 PS C:\Users\Love\OneDrive\Desktop\AI.AC> C:\Users\Love\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Love/OneDrive/Desktop/AI.AC/lab 1.py"
59 10 -> 20 -> 30
60 PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Singly Linked List is a dynamic data structure where elements (nodes) are connected using pointers. Linked Lists are useful when frequent insertions and deletions are required, as they do not require shifting elements like arrays.

Task Description #4: Binary Search Tree (BST)

Task: Use AI to create a BST with insert and in-order traversal methods.

Prompt: Create a Binary Search Tree in Python with recursive insert and inorder traversal methods. Include proper class structure and documentation.

```
150 ## TASK-4: Create a Binary Search Tree in Python with a nested Node class. Implement recursive insert and in-order traversal
151 methods following BST properties. Add proper docstrings.
152 class Node:
153     def __init__(self, data):
154         self.data = data
155         self.left = None
156         self.right = None
157
158 class BinarySearchTree:
159     def __init__(self):
160         self.root = None
161
162     def insert(self, data):
163         if self.root is None:
164             self.root = Node(data)
165             print(f"{data} inserted as root of the BST.")
166         else:
167             self._insert_recursive(self.root, data)
168
169     def _insert_recursive(self, node, data):
170         if data < node.data:
171             if node.left is None:
172                 node.left = Node(data)
173                 print(f"{data} inserted to the left of {node.data}.")
174             else:
175                 self._insert_recursive(node.left, data)
176         else:
177             if node.right is None:
178                 node.right = Node(data)
179                 print(f"{data} inserted to the right of {node.data}.")
180             else:
181                 self._insert_recursive(node.right, data)
182
183     def in_order_traversal(self):
184         elements = []
185         self._in_order_recursive(self.root, elements)
186         print(f"In-order Traversal: " + " ".join(map(str, elements)))
187
188 if __name__ == "__main__":
189     bst = BinarySearchTree()
190     while True:
191         print("\n1. Insert")
192         print("2. In-order Traversal")
193         print("3. Exit")
194         choice = input("Enter your choice: ")
195         if choice == '1':
196             value = input("Enter value to insert: ")
197             bst.insert(value)
198         elif choice == '2':
199             bst.in_order_traversal()
200         elif choice == '3':
201             print("Exiting program...")
202             break
203         else:
204             print("Invalid choice! Try again.")
```

```
182 class BinarySearchTree:
183     def in_order_traversal(self):
184         elements = []
185         self._in_order_recursive(self.root, elements)
186         print("In-order Traversal: " + " ".join(map(str, elements)))
187
188     def _in_order_recursive(self, node, elements):
189         if node:
190             self._in_order_recursive(node.left, elements)
191             elements.append(node.data)
192             self._in_order_recursive(node.right, elements)
193
194 bst = BinarySearchTree()
195 while True:
196     print("\n1. Insert")
197     print("2. In-order Traversal")
198     print("3. Exit")
199     choice = input("Enter your choice: ")
200     if choice == '1':
201         value = input("Enter value to insert: ")
202         bst.insert(value)
203     elif choice == '2':
204         bst.in_order_traversal()
205     elif choice == '3':
206         print("Exiting program...")
207         break
208     else:
209         print("Invalid choice! Try again.")
```

OUTPUT:

```
PS C:\Users\sarik\OneDrive\Desktop\AI ASSISTED CODING> & C:\Users\sarik\AppData\Local\Python\pythoncore-3.14-64\python.exe "C:\Users\sarik\OneDrive\Desktop\AI ASSISTED CODING\ASSIGN-11-1.py"
1. Insert
2. In-order Traversal
3. Exit
Enter your choice: 1
Enter value to insert: 11
11 inserted as root of the BST.

1. Insert
2. In-order Traversal
3. Exit
Enter your choice: 1
Enter value to insert: 14
14 inserted to the right of 11.

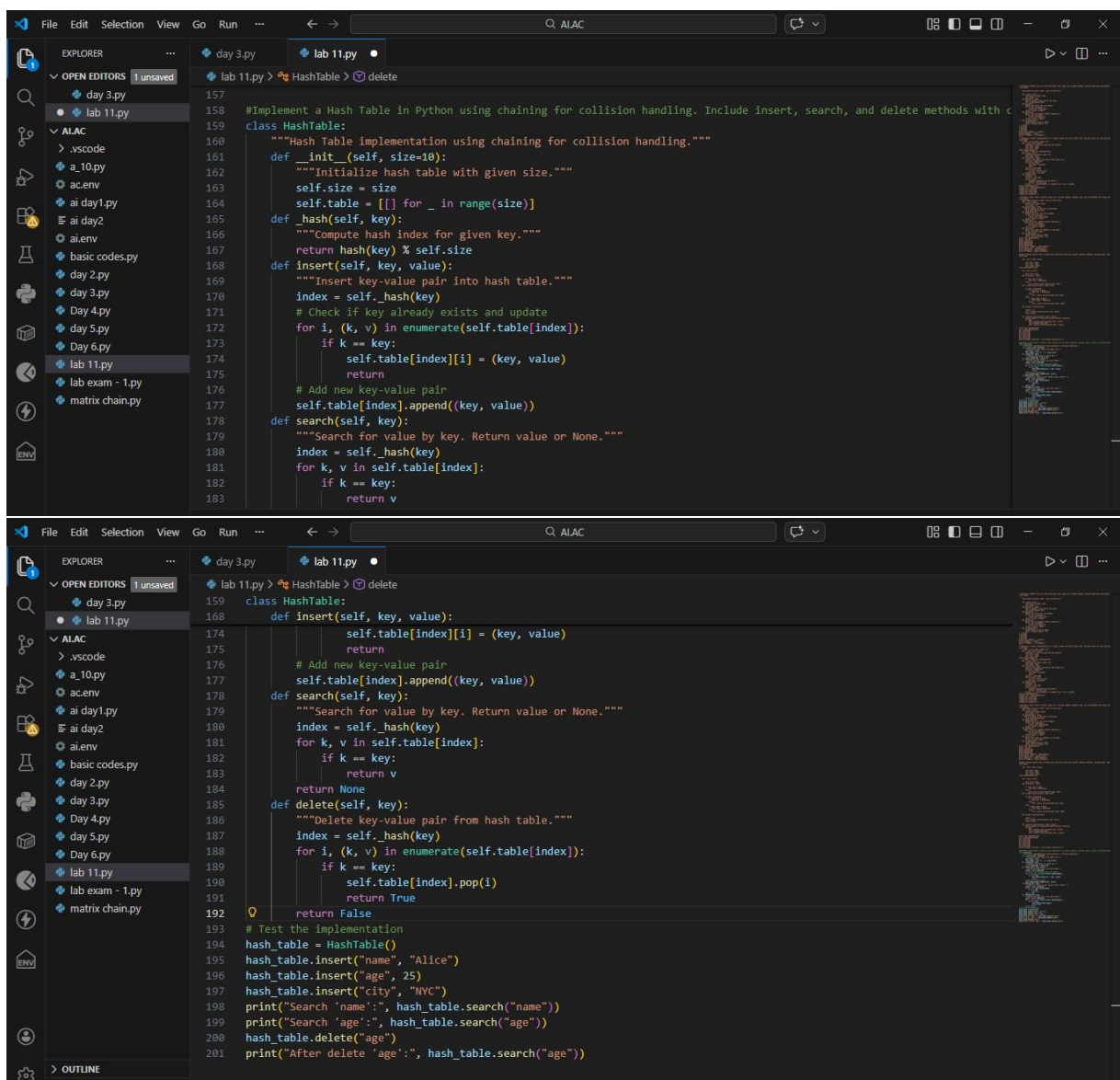
1. Insert
2. In-order Traversal
3. Exit
Enter your choice: 2
```

Explanation: A Binary Search Tree is a hierarchical data structure where the left child contains smaller values and the right child contains larger values than the root. This property makes searching, insertion, and deletion efficient.

Task Description #5: Hash Table

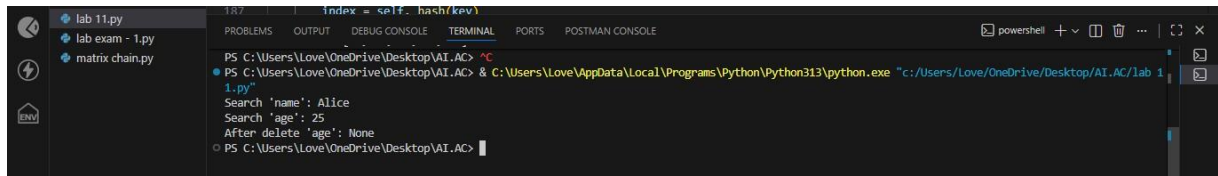
Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Prompt: Implement a Hash Table in Python using chaining for collision handling. Include insert, search, and delete methods with comments.



```
157
158 #Implement a Hash Table in Python using chaining for collision handling. Include insert, search, and delete methods with c
159 class HashTable:
160     """Hash Table implementation using chaining for collision handling."""
161     def __init__(self, size=10):
162         """Initialize hash table with given size."""
163         self.size = size
164         self.table = [[] for _ in range(size)]
165     def _hash(self, key):
166         """Compute hash index for given key."""
167         return hash(key) % self.size
168     def insert(self, key, value):
169         """Insert key-value pair into hash table."""
170         index = self._hash(key)
171         # Check if key already exists and update
172         for i, (k, v) in enumerate(self.table[index]):
173             if k == key:
174                 self.table[index][i] = (key, value)
175                 return
176         # Add new key-value pair
177         self.table[index].append((key, value))
178     def search(self, key):
179         """Search for value by key. Return value or None."""
180         index = self._hash(key)
181         for k, v in self.table[index]:
182             if k == key:
183                 return v
184
185
186
187
188
189
190
191
192
193 # Test the implementation
194 hash_table = HashTable()
195 hash_table.insert("name", "Alice")
196 hash_table.insert("age", 25)
197 hash_table.insert("city", "NYC")
198 print("Search 'name':", hash_table.search("name"))
199 print("Search 'age':", hash_table.search("age"))
200 hash_table.delete("age")
201 print("After delete 'age':", hash_table.search("age"))
```


OUTPUT:



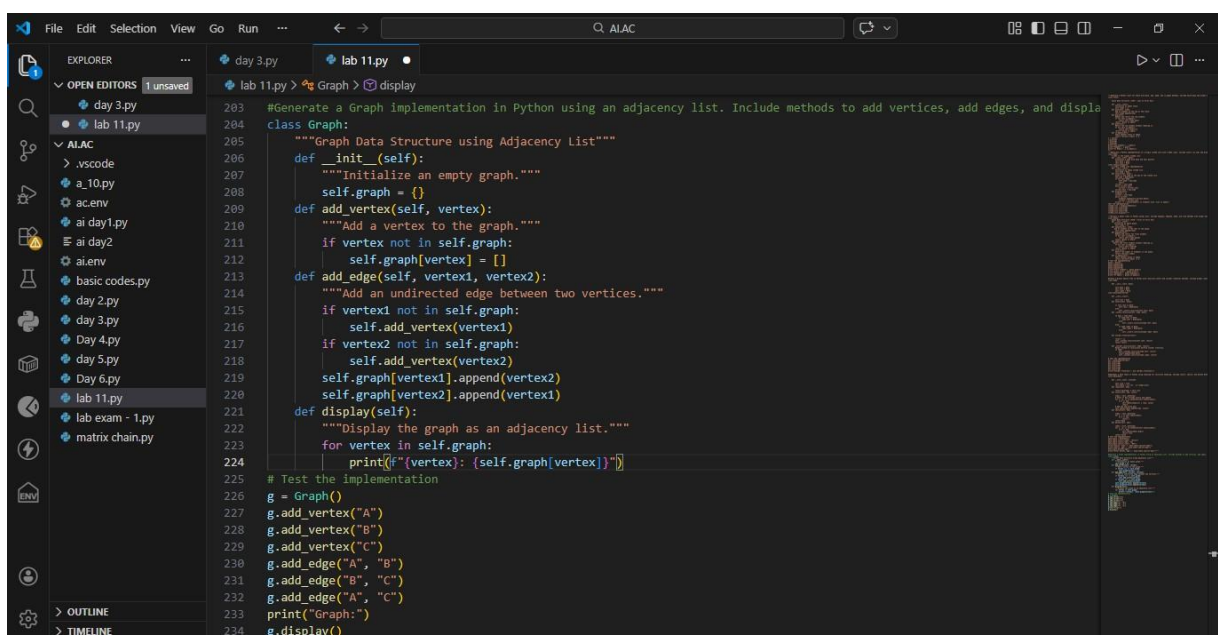
```
187         index = self.hash(key)
PS C:\Users\Love\OneDrive\Desktop\AI.AC> ^C
PS C:\Users\Love\OneDrive\Desktop\AI.AC> & C:\Users\Love\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Love/OneDrive/Desktop/AI.AC/lab 11.py"
Search 'name': Alice
Search 'age': 25
After delete 'age': None
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Hash Table stores data in key-value pairs using a hash function to compute an index. It provides fast average-case time complexity for search, insertion, and deletion operations.

Task Description #6: Graph Representation

Task: Use AI to implement a graph using an adjacency list.

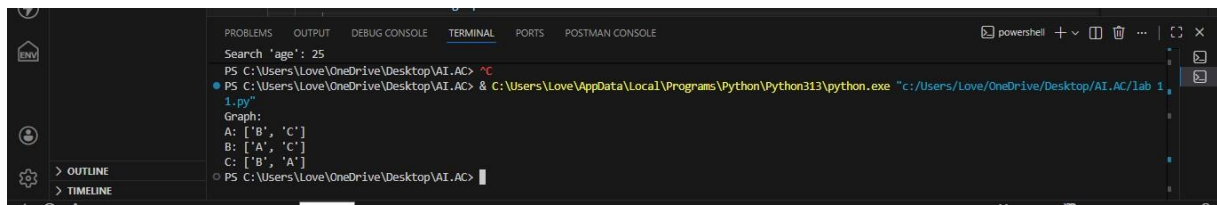
Prompt: Generate a Graph implementation in Python using an adjacency list. Include methods to add vertices, add edges, and display the graph.



```
File Edit Selection View Go Run ... ALAC
EXPLORER
  OPEN EDITORS 1 unsaved
    day 3.py
    lab 11.py
  ALAC
    .vscode
    a_10.py
    ac.env
    ai day1.py
    ai day2
    ai.env
    basic codes.py
    day 2.py
    day 3.py
    Day 4.py
    day 5.py
    Day 6.py
    lab 11.py
    lab exam - 1.py
    matrix chain.py
  OUTLINE
  TIMELINE

lab 11.py > Graph > display
283 #Generate a Graph implementation in Python using an adjacency list. Include methods to add vertices, add edges, and display the graph.
284 class Graph:
285     """Graph Data Structure using Adjacency List"""
286     def __init__(self):
287         """Initialize an empty graph."""
288         self.graph = {}
289     def add_vertex(self, vertex):
290         """Add a vertex to the graph."""
291         if vertex not in self.graph:
292             self.graph[vertex] = []
293     def add_edge(self, vertex1, vertex2):
294         """Add an undirected edge between two vertices."""
295         if vertex1 not in self.graph:
296             self.add_vertex(vertex1)
297         if vertex2 not in self.graph:
298             self.add_vertex(vertex2)
299         self.graph[vertex1].append(vertex2)
300         self.graph[vertex2].append(vertex1)
301     def display(self):
302         """Display the graph as an adjacency list."""
303         for vertex in self.graph:
304             print(f"{vertex}: {self.graph[vertex]}")
305 # Test the implementation
306 g = Graph()
307 g.add_vertex("A")
308 g.add_vertex("B")
309 g.add_vertex("C")
310 g.add_edge("A", "B")
311 g.add_edge("B", "C")
312 g.add_edge("A", "C")
313 print("Graph:")
314 g.display()
```

Output:

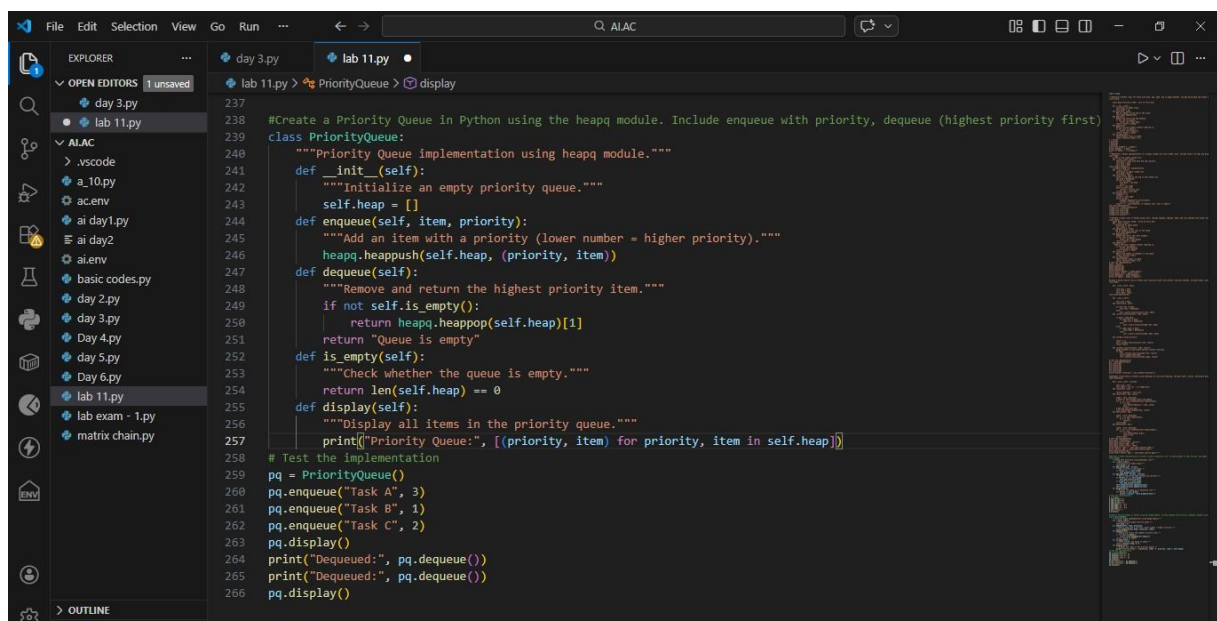


Explanation: A Graph is a non-linear data structure used to represent relationships between entities. It consists of vertices (nodes) and edges (connections).

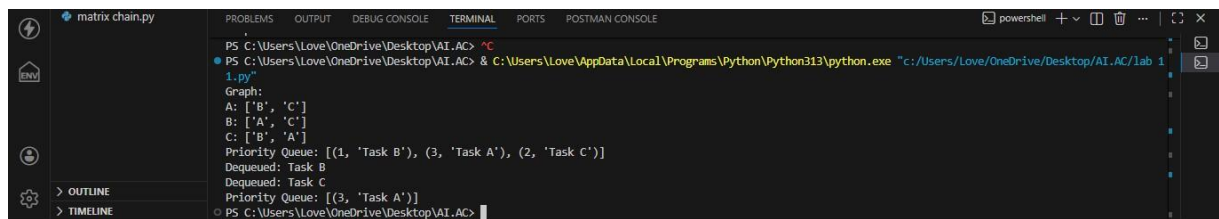
Task Description #7: Priority Queue

Task: Use AI to implement a priority queue using Python's heap module.

Prompt: Create a Priority Queue in Python using the heapq module. Include enqueue with priority, dequeue (highest priority first), and display methods.



Output:



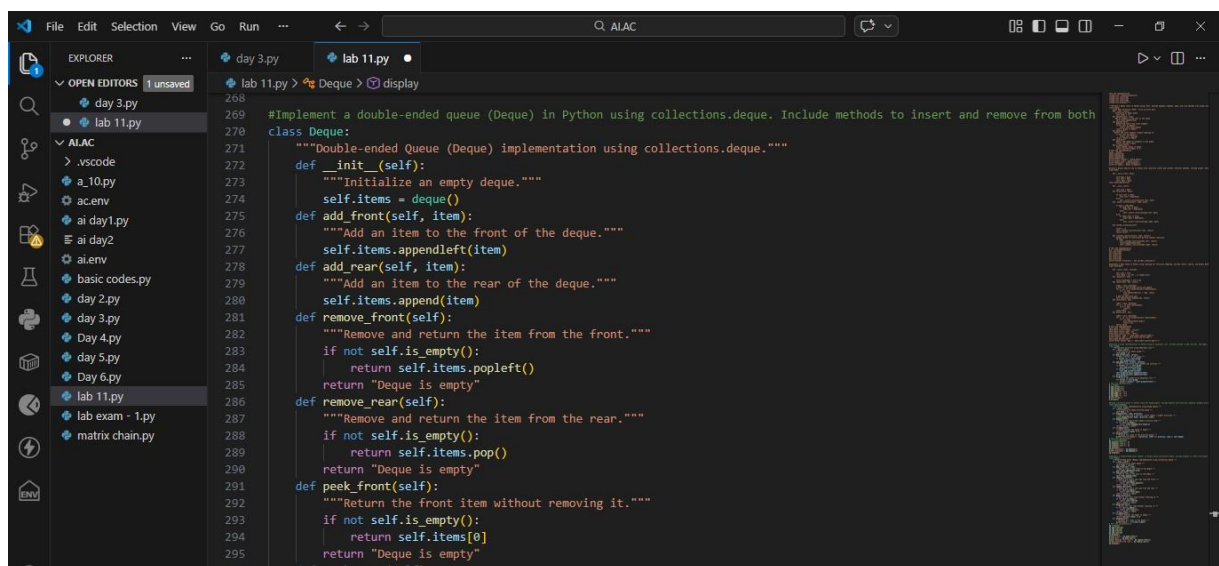
```
PS C:\Users\Love\OneDrive\Desktop\AI.AC> ^C
PS C:\Users\Love\OneDrive\Desktop\AI.AC> & C:\Users\Love\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Love/OneDrive/Desktop/AI.AC/lab 1
1.py"
Graph:
A: ['B', 'C']
B: ['A', 'C']
C: ['B', 'A']
Priority Queue: [(1, 'Task B'), (3, 'Task A'), (2, 'Task C')]
Dequeued: Task B
Dequeued: Task C
Priority Queue: [(3, 'Task A')]
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Priority Queue is a special type of queue where elements are removed based on priority rather than order of insertion. Higher priority elements are processed first. It is typically implemented using a heap for efficiency.

Task Description #8 – Deque

Task: Use AI to implement a double-ended queue using `collections.deque`.

Prompt: Implement a double-ended queue (Deque) in Python using `collections.deque`. Include methods to insert and remove from both ends with documentation.



```
268
269
270 #Implement a double-ended queue (Deque) in Python using collections.deque. Include methods to insert and remove from both
class Deque:
271     """Double-ended Queue (Deque) implementation using collections.deque."""
272     def __init__(self):
273         """Initialize an empty deque."""
274         self.items = deque()
275     def add_front(self, item):
276         """Add an item to the front of the deque."""
277         self.items.appendleft(item)
278     def add_rear(self, item):
279         """Add an item to the rear of the deque."""
280         self.items.append(item)
281     def remove_front(self):
282         """Remove and return the item from the front."""
283         if not self.is_empty():
284             return self.items.popleft()
285         return "Deque is empty"
286     def remove_rear(self):
287         """Remove and return the item from the rear."""
288         if not self.is_empty():
289             return self.items.pop()
290         return "Deque is empty"
291     def peek_front(self):
292         """Return the front item without removing it."""
293         if not self.is_empty():
294             return self.items[0]
295         return "Deque is empty"
296     def peek_rear(self):
297         """Return the rear item without removing it."""
```



```
270 class Deque:
286 def remove_rear(self):
290     return "Deque is empty"
291
292 def peek_front(self):
293     """Return the front item without removing it."""
294     if not self.is_empty():
295         return self.items[0]
296     return "Deque is empty"
297
298 def peek_rear(self):
299     """Return the rear item without removing it."""
300     if not self.is_empty():
301         return self.items[-1]
302     return "Deque is empty"
303
304 def is_empty(self):
305     """Check whether the deque is empty."""
306     return len(self.items) == 0
307
308 def display(self):
309     """Display all items in the deque."""
310     print("Deque:", list(self.items))
311
312 # Test the implementation
313 dq = Deque()
314 dq.add_front(10)
315 dq.add_rear(20)
316 dq.add_front(5)
317 dq.add_rear(30)
318 dq.display()
319 print("Front:", dq.peek_front())
320 print("Rear:", dq.peek_rear())
321 print("Removed from front:", dq.remove_front())
322 print("Removed from rear:", dq.remove_rear())
323 dq.display()
```

Output:

```
Priority Queue: [(3, 'Task A')]
Deque: [5, 10, 20, 30]
Front: 5
Rear: 30
Removed from front: 5
Removed from rear: 30
Deque: [10, 20]
PS C:\Users\Love\OneDrive\Desktop\AI.AC>
```

Explanation: A Deque (Double Ended Queue) allows insertion and deletion of elements from both the front and rear ends.